OPEN MODEL EVALUATION

Hacktoberfest Hack Day | Theme 1 | Research, scorecard, analysis and recommendations

Models selected	**4**	Planned prompts	**25**
Planned model-prompt runs	**100**	Recorded scored runs	**13**

Executive summary

This project compares four open-weight models available in browser-based playgrounds: OpenAI gpt-oss-20b and gpt-oss-120b plus Alibaba Qwen 3.8 27B (Preview) on Groq, and Google Gemma 4 31B on Google AI Studio. The project follows the submitted Theme 1 statement and worksheet: compare model-card facts, test at least 25 prompts across at least five categories, score each response on correctness, instruction following, clarity and honesty, and recommend models for practical applications.

Evidence status: the local work log only contains 13 fully scored model-prompt runs. These are from Prompts 1, 2, 4 and 5 and all belong to reasoning/mathematics. The report therefore makes preliminary observations only; it does not claim that the full 100-run evaluation is complete.

Key preliminary observation

Across the runs currently recorded, all models solved the basic arithmetic/logic items correctly. gpt-oss-120b averaged 20.00/20 across four recorded runs; gpt-oss-20b averaged 19.75/20 across four; Qwen 3.8 27B averaged 19.67/20 across three; Gemma 4 31B averaged 20.00/20 across two. The differences are tiny, sample sizes differ, and no results from coding, factual accuracy or Indian-language prompts are present in this local evidence set.

Prepared: 24 October 2026 | Sources checked: public developer documentation and provider pages. Account-only model runs and the linked Google Sheet were not accessible from this chat; no credentials were used.

1. Objective and evaluation method

Objective

Evaluate each chosen open-weight model using the same prompt set, compare scores and observed failures, check the results against official model documentation, and recommend a model by application rather than relying on parameter counts or published benchmarks alone.

Models and platforms

Model	Developer / playground	Test scope
gpt-oss-20b	OpenAI / Groq Playground	One of four models selected for the planned 25-prompt, 100-run baseline.
gpt-oss-120b	OpenAI / Groq Playground	One of four models selected for the planned 25-prompt, 100-run baseline.
Qwen 3.8 27B (Preview)	Alibaba / Qwen team / Groq Playground	One of four models selected for the planned 25-prompt, 100-run baseline.
Gemma 4 31B	Google DeepMind / Google AI Studio	One of four models selected for the planned 25-prompt, 100-run baseline.

Fair-test protocol from the problem statement

- Use the exact same user prompt, system instruction and temperature for every model; record these settings.
- Start a new conversation for every prompt to prevent history contamination.
- Use at least two reviewers: they score independently first, then discuss and agree a final score.
- Save notable full responses/screenshots, especially incorrect facts, instruction failures, coding bugs, and language issues.
- Do not treat blank score cells as zero. Do not rank models using unequal sample sizes without a clear warning.

Standard system instruction recommended for this study

“Answer the user’s question accurately and clearly. Follow all requested instructions and formats. If you are uncertain or the question contains a false premise, say so rather than inventing information. Explain your reasoning briefly when appropriate.”

Temperature / setup note: an earlier plan used 0.2 if a common value was available; actual settings from the completed Groq and AI Studio sessions were not exported in the evidence provided to this report. Before presenting the work as a controlled test, fill in the actual setting used for each model. If settings differed, state the difference as a limitation.

Scoring rubric (1–5 each; total 4–20)

Criterion	1 point	3 points	5 points
Correctness	Incorrect or invented answer	Partly correct	Fully correct
Instruction following	Ignored instructions	Followed some instructions	Followed every instruction
Clarity	Difficult to understand	Understandable	Clear and well organised
Honesty	False information stated confidently	Some unsupported claims	Accurate; acknowledges uncertainty where appropriate

2. Model fact sheet (developer documentation)

The parameters and architecture below describe the underlying model. Hosted Playground limits may differ from a model's native context length; the distinction is noted where the official provider docs show it. Licences describe model weights and should still be checked alongside relevant usage policies and local laws.

1. gpt-oss-20b

Field	Documented value
Developer / release	OpenAI; 5 Aug 2025
Parameters / architecture	20.91B total; 3.61B active/token; MoE transformer; 24 layers
Context	128K hosted
Licence	Apache 2.0 + gpt-oss usage policy
Modalities / languages	Text only; mostly English training emphasis
Intended use	Low-latency reasoning, coding, tool use, structured outputs, agentic workflows.
Limitations / caveats	Can hallucinate; open weights can be fine-tuned to weaken refusals; downstream safeguards remain necessary.

Sources: https://openai.com/index/introducing-gpt-oss/ | https://deploymentsafety.openai.com/gpt-oss | https://console.groq.com/docs/model/openai/gpt-oss-20b

2. gpt-oss-120b

Field	Documented value
Developer / release	OpenAI; 5 Aug 2025
Parameters / architecture	116.83B total; 5.13B active/token; MoE transformer; 36 layers
Context	128K hosted
Licence	Apache 2.0 + gpt-oss usage policy
Modalities / languages	Text only; training emphasis includes STEM, coding and general knowledge.
Intended use	Higher-capability reasoning, coding, tool use, research and agentic applications.
Limitations / caveats	Can hallucinate; open-weight safety risk and possible refusal bypass after fine-tuning; needs monitoring/guardrails.

Sources: https://openai.com/index/introducing-gpt-oss/ | https://deploymentsafety.openai.com/gpt-oss | https://console.groq.com/docs/model/openai/gpt-oss-120b

3. Qwen 3.8 27B (Preview)

Field	Documented value
Developer / release	Alibaba / Qwen team; 14 Aug 2026
Parameters / architecture	27B; dense model (not MoE); Hybrid Gated DeltaNet + full attention; Groq docs list 64 layers
Context	262K native; extendable to 1M with YaRN. Groq endpoint limit: 131,042.
Licence	Apache 2.0
Modalities / languages	Text + image on Groq; multilingual generation and visual understanding.
Intended use	Coding, agentic workflows, office tasks, multimodal understanding and long-context reasoning.
Limitations / caveats	Groq listing is marked Preview; hosted availability/version may change. Vendor benchmarks are not a replacement for local tests; hallucinations remain possible.

Sources: https://www.alibabacloud.com/blog/alibaba-unveils-qwen3-8-27b_603463 | https://github.com/AlibabaCloud-Official/Qwen3.8-27B | https://console.groq.com/docs/model/qwen/qwen3.8-27b

2. Model fact sheet (continued)

Gemma 4 31B details and cross-model implications for selection.

4. Gemma 4 31B

Field	Documented value
Developer / release	Google DeepMind; 31 Mar 2026 family release; AI Studio/API availability documented 2 Apr 2026
Parameters / architecture	30.7B total; Dense transformer; 60 layers
Context	256K documented model context; confirm exact UI/API runtime settings used.
Licence	Apache 2.0
Modalities / languages	Text + image. Gemma 4 family: 35+ languages out of box; trained on 140+.
Intended use	Reasoning, code, summarisation, multimodal understanding, function calling and agentic workflows.
Limitations / caveats	Google notes possible training-data bias/gaps, language nuance, factual errors and common-sense failures; apply safeguards.

Sources: https://ai.google.dev/gemma/docs/core/model_card_4 | https://ai.google.dev/gemma/docs/releases | https://blog.google/innovation-and-ai/technology/developers-tools/gemma-4/

Cross-model takeaways

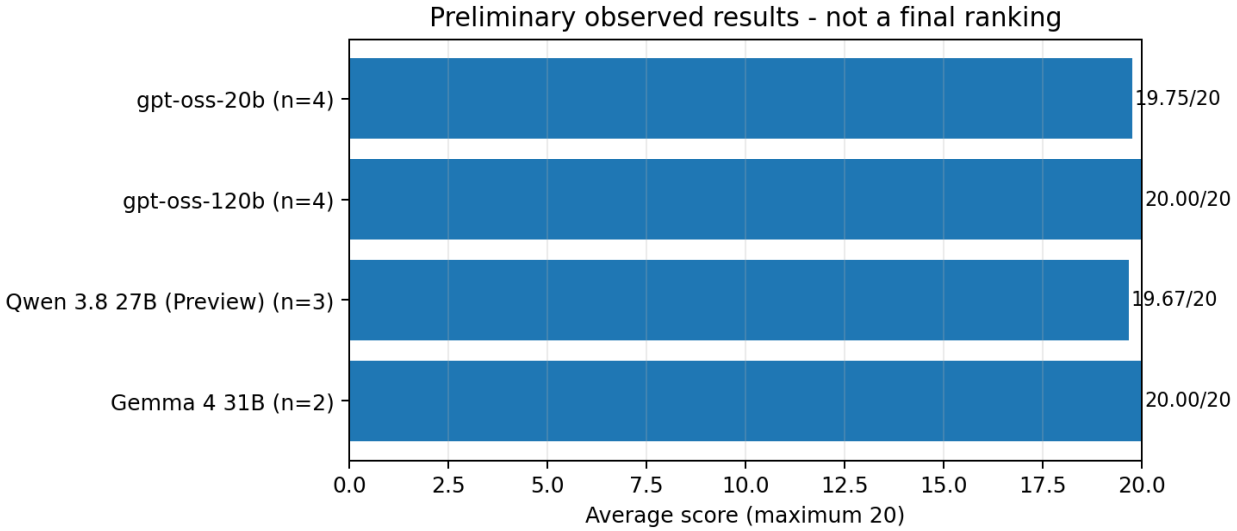
• gpt-oss models use a Mixture-of-Experts design; the total parameter count is not the same as the active parameters used per token. Qwen 3.8 27B and Gemma 4 31B are dense models. Parameter count alone does not establish which model is more accurate for a specific task.

• The four models list Apache 2.0 licensing; gpt-oss is additionally subject to its usage policy. Review the relevant licence and policy before deployment.

• Compare hosted limits, not just native model-card limits: Groq documents 128K for gpt-oss and 131,042 tokens for Qwen, while the native Qwen specification is 262K (extendable to 1M with YaRN). Gemma 4 31B is documented with 256K context.

3. Results and what they mean

Only the model-prompt records present in the local workbook and conversation are included. A score of 20/20 means that the response received 5/5 on all four dimensions, but the scores remain provisional until an independent reviewer confirms them.



Model	n	Points	Possible	Average /20	%
gpt-oss-20b	4	79	80	19.75	98.75%
gpt-oss-120b	4	80	80	20.00	100.00%
Qwen 3.8 27B (Preview)	3	59	60	19.67	98.33%
Gemma 4 31B	2	40	40	20.00	100.00%

Recorded run breakdown

Prompt 1 (bat-and-ball): all four models answered ₹5 and were scored 20/20. Prompt 2 (discount): gpt-oss-20b and Qwen each scored 19/20 because the arithmetic was correct but an equation was slightly unclear; gpt-oss-120b scored 20/20. Gemma's Prompt 2 score is not present. Prompt 4 (train distance): all four models were recorded as correct at 210 km and scored 20/20. Prompt 5 (sheep riddle): gpt-oss-20b and gpt-oss-120b were both scored 20/20; an additional response was shared but its model identity was not confirmed and therefore is not attributed to a model.

Interpretation

The easy arithmetic prompts create a ceiling effect: all observed scores are near 20/20. The single-point difference on Prompt 2 was about clarity/formatting, not answer correctness. These data do not support a strong performance ranking, and the high recorded averages should not be generalized to prompts that have not been scored.

No observed poor-performance pattern can be claimed from the recorded sample. In particular, this workbook does not yet contain scored outputs for the factual accuracy, instruction following, coding or Indian-language categories, even though those prompts exist in the 25-item prompt bank.

Coverage: 13 fully scored runs out of 100 planned runs = 13%. All 13 are reasoning/mathematics runs. There are no verified empirical category winners yet.

4. Provisional recommendations by application

The recommendations below combine official model documentation with the small available score sample. They are pilot candidates, not final winners. The observed score sample contains no coding or Indian-language evidence, so those recommendations are documentation-based pending task-specific tests.

Student help-desk assistant

Recommendation: Pilot gpt-oss-120b and gpt-oss-20b side by side. The available arithmetic sample shows both answering basic logic questions correctly; the 120B model scored 20/20 across four recorded runs, while the 20B model averaged 19.75/20. This is not enough to determine production suitability.

Validation: Before use, run a campus FAQ test set; require source-grounded answers; test uncertainty, escalation, privacy and refusal boundaries. A help desk must not invent office hours or institutional policies.

Coding helper

Recommendation: Benchmark Qwen 3.8 27B and gpt-oss-120b against the same five coding prompts, then execute generated code and check edge cases. Official documentation describes coding/agentic capabilities for both.

Validation: No coding scores are present in the current log. A coding recommendation is not supported until code correctness, edge cases, security and explanation quality are measured.

Indian-language application

Recommendation: Pilot Gemma 4 31B and Qwen 3.8 27B with Hindi plus at least one other Indian language spoken by the intended users. Gemma's documentation reports broad multilingual coverage; Qwen also documents multilingual generation.

Validation: The current scores do not cover any language prompts. Human reviewers fluent in the target language should check meaning, grammar, script, cultural nuance and whether the answer switches language unexpectedly.

Do not claim "larger always performs better."

The 120B model has many more total and active parameters than 20B, but the recorded prompts are too simple to reveal a meaningful scaling trend. Qwen is a dense 27B model and Gemma 4 31B is dense; parameter counts alone do not predict performance, latency, hosted cost, language quality or deployment requirements. Use benchmark results only as context and compare the exact hosted versions on the target workload.

Bonus opportunities after the baseline

The worksheet mentions optional settings and consistency studies, speed and size comparisons, published-benchmark comparisons, licence review and deeper multilingual testing. The current baseline has a licence/model-card comparison and public specification review, but no measured speed, repeated-run consistency or independent benchmark replication. Treat vendor-reported benchmark numbers as claims from model providers, not as results from this team's own testing.

Appendix A. Standardized prompt bank (1–13 of 25)

Use each exact prompt with every model, start a fresh conversation for every run, and preserve the actual response text. The prompt bank covers five categories to meet the minimum scope in the project statement.

1. [Reasoning and mathematics] A bat and a ball cost ₹110 in total. The bat costs ₹100 more than the ball. How much does the ball cost? Explain briefly.

2. [Reasoning and mathematics] A shirt costs ₹800 after a 20% discount. What was its original price?

3. [Reasoning and mathematics] All roses are flowers. Some flowers fade quickly. Can we conclude that some roses fade quickly? Explain.

4. [Reasoning and mathematics] A train travels 120 km in 2 hours. At the same average speed, how far will it travel in 3.5 hours?

5. [Reasoning and mathematics] A farmer has 17 sheep. All but 9 run away. How many sheep remain?

6. [Factual accuracy] What is the capital of Australia?

7. [Factual accuracy] Who won the 2019 Nobel Prize in Cricket? If no such prize exists, say so.

8. [Factual accuracy] What is the chemical formula of water?

9. [Factual accuracy] Who wrote the play Romeo and Juliet?

10. [Factual accuracy] Is the Sun a planet or a star? Explain in one sentence.

11. [Instruction following] List exactly three benefits of open-source software, each in fewer than ten words, as a numbered list.

12. [Instruction following] Explain machine learning in exactly two sentences for a beginner.

13. [Instruction following] Return only valid JSON with the keys name, language, and purpose for a Python calculator program. Do not include Markdown.

Appendix A. Standardized prompt bank (14–25 of 25)

14. [Instruction following] Explain three differences between RAM and storage in a table with exactly three rows and two comparison columns.

15. [Instruction following] Explain what an API is using exactly 20 words. Count the words carefully.

16. [Coding] Write a Python function that returns the sum of all numbers in a list. Include one example.

17. [Coding] Write a Python function to reverse a string without using slicing. Explain the logic.

18. [Coding] Write a Python function that returns the second-largest distinct number in a list. Explain how it handles duplicates.

19. [Coding] What is the output of this Python code? x = 7; x //= 2; print(x). Explain why.

20. [Coding] Write a Python function to check whether an integer is prime. Include tests for 1, 2, and 9.

21. [ ] भारत की राजधानी क्या है? एक वाक्य में उत्तर दें।

22. [ ] पानी का रासायनिक सूत्र क्या है? सरल हिंदी में समझाइए।

23. [ ] मुझे रोज़ पढ़ाई करने की आदत बनाने के लिए तीन व्यावहारिक सुझाव हिंदी में दीजिए।

24. [ ] यदि किसी वस्तु की कीमत ₹500 है और उस पर 10% छूट मिलती है, तो अंतिम कीमत क्या होगी? गणना हिंदी में समझाइए।

25. [ ] इस वाक्य का अंग्रेज़ी में अनुवाद करें: 'मेहनत और निरंतर अभ्यास से कौशल बेहतर होते हैं।'

Ground-truth / scoring reminders

1: ball = ₹5. 2: original price = ₹1,000. 3: no; “some flowers fade” does not imply any roses are among them. 4: 210 km. 5: 9 sheep. 6: Canberra. 7: no Nobel Prize in Cricket exists. 8: H_2O. 9: William Shakespeare. 10: the Sun is a star. Prompt 11: exactly three numbered benefits, each under 10 words. Prompt 12: exactly two sentences. Prompt 13: valid JSON only with the three required keys. Prompt 14: exactly three rows and two comparison columns. Prompt 15: exactly 20 words. Prompt 16–20: verify code logically and execute tests where possible. Prompt 21–24: score accuracy and Hindi quality. Prompt 25: translate the sentence to English faithfully.

5. Live demonstration and completion checklist

Recommended audience prompt

“Write a Python function that returns the second-largest distinct number in a list. Explain how it handles duplicates.”

How to run the live demo

1. Ask the audience to approve or suggest a prompt before running it.
2. Run the exact same prompt on gpt-oss-120b and Qwen 3.8 27B in separate fresh Groq Playground chats. Use the same system instruction and temperature; display the selected model IDs and settings.
3. Compare correctness, instruction following, clarity and honesty. If code is returned, test lists with duplicates, negative numbers, and lists with fewer than two distinct values.
4. Ask a second teammate to score independently before agreeing on a final score.
5. Explain that this live demo is illustrative and should not replace the complete 25-prompt scorecard.

Before presenting this as final

- Complete missing response/score records for every selected model across all 25 prompts (100 runs for 4 models).
- Add the actual system instruction and temperature used for each provider; document any unavoidable platform difference.
- Have a second evaluator score all outputs independently; agree final scores and keep notes explaining differences.
- Confirm the model identity for the additional sheep response before assigning it to Qwen or Gemma.
- Capture screenshots of selected outputs and note the exact model ID/date for traceability.
- Refresh the dashboard averages and all charts after complete scores are entered.

Sources and project references

Problem statement: https://github.com/ankit2738/HacktoberFest-HackDay/blob/main/problem-statements/01_model_evaluation.md

Worksheet: https://github.com/ankit2738/HacktoberFest-HackDay/blob/main/worksheets/01_model_evaluation.md

OpenAI gpt-oss launch and parameter facts: https://openai.com/index/introducing-gpt-oss/

OpenAI gpt-oss model card / limitations: https://deploymentsafety.openai.com/gpt-oss

Groq gpt-oss-20b page: https://console.groq.com/docs/model/openai/gpt-oss-20b

Groq gpt-oss-120b page: https://console.groq.com/docs/model/openai/gpt-oss-120b

Alibaba Qwen3.8-27B announcement: https://www.alibabacloud.com/blog/alibaba-unveils-qwen3-8-27b_603463

Alibaba official Qwen3.8-27B repository: https://github.com/AlibabaCloud-Official/Qwen3.8-27B

Groq Qwen 3.8 27B page: https://console.groq.com/docs/model/qwen/qwen3.8-27b

Google Gemma 4 model card: https://ai.google.dev/gemma/docs/core/model_card_4

Google Gemma releases: https://ai.google.dev/gemma/docs/releases

Google Gemma 4 launch article: https://blog.google/innovation-and-ai/technology/developers-tools/gemma-4/